Data Flow Diagram

Purpose

This diagram shows the main information paths in trvale-book: the traveller's preferences, request validation, itinerary generation, budget calculation, persistence and printing.

Level 1 Data Flow Diagram

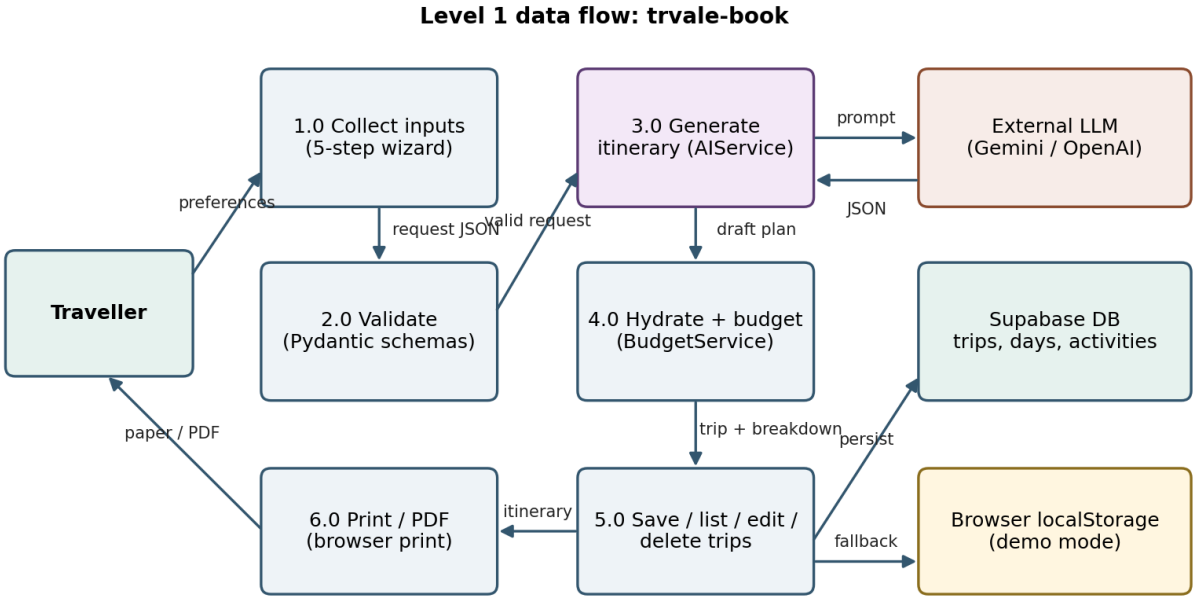


Figure 1. Data flow between the traveller, the API, the AI service, the database and the browser.

Explanation of the flow

1. The traveller completes the five-step wizard in the React frontend. The frontend sends the preferences as JSON to the backend generate endpoint.
2. The backend validates the request with Pydantic schemas (for example, the end date must not be earlier than the start date, the budget must be at least 50 and the traveller count between 1 and 50).
3. The AI service builds a prompt and calls Google Gemini or OpenAI. The JSON reply is parsed, given identifiers and normalized. If no key is configured or the call fails, a heuristic generator creates the plan instead.
4. The budget service totals lodging, food, activity and transit costs, compares them with the target budget, and adds warnings when the plan is over budget or leaves a large remainder.
5. Trips are listed, opened, edited and deleted through the trips endpoints. When Supabase is configured the trip record is inserted into the database; in demo mode, data lives in server memory and in the browser's localStorage.
6. The Print / PDF action opens the browser print dialog; a print stylesheet formats the itinerary.

Data stores, controls and limitations

Supabase (PostgreSQL with row-level security) is the intended persistent store for profiles, trips, preferences, days and activities. Gemini and OpenAI keys and the Supabase service credentials belong in backend environment variables only. Generated plans and costs are estimates. The current backend keeps generated trips in an in-memory dictionary and inserts only the trip row into Supabase on creation, so full multi-user persistence of days and activities is future work.